Why These Losses?

A probabilistic refresher for Deep Learning

Prof. Nipun Batra

ES 667 · Deep Learning · IIT Gandhinagar

Three familiar equations

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Why exactly these three expressions? Today: they are not arbitrary.

The map for today

V VISUAL

Two probabilistic assumptions, two halves of the objective:

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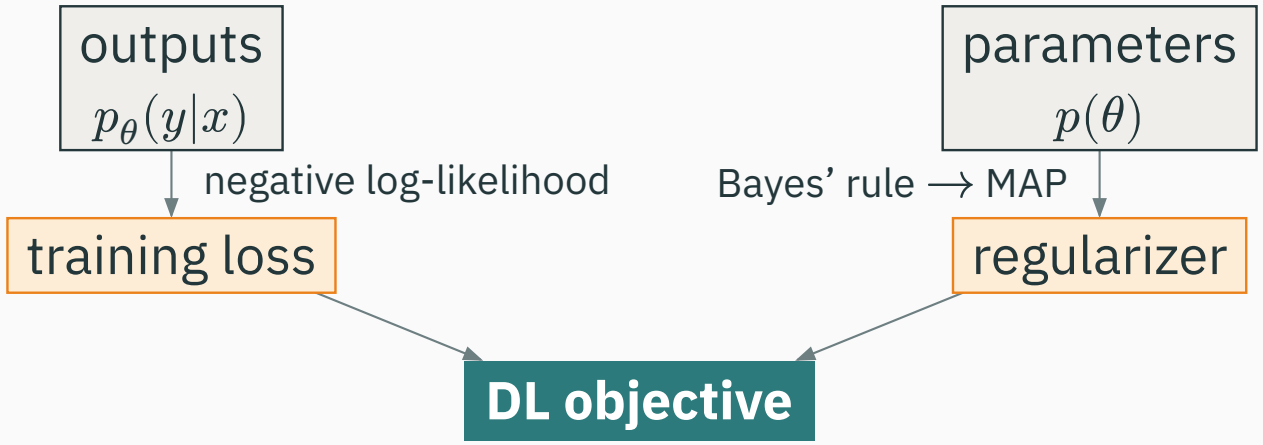

Minimal probability revision

A model should describe uncertainty

$\hat{y} = f_\theta(x)$

a single number

$Y \mid X = x \sim p_\theta(\cdot \mid x)$

a full distribution

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A neural network parameterizes a probability distribution.

Density is not probability

V VISUAL

For a continuous Y, probability comes from **area**:

$$P(a \leq Y \leq b) = \int_a^b p(y)\,\mathrm{d}y$$

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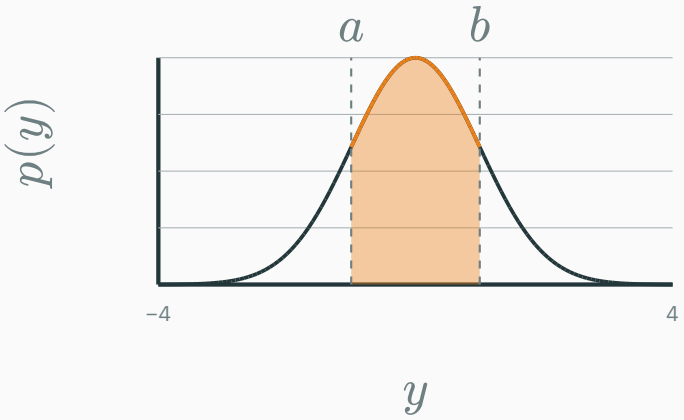
$p(y) \neq P(Y = y)$ — a density can **exceed 1**; only *integrals* are probabilities.



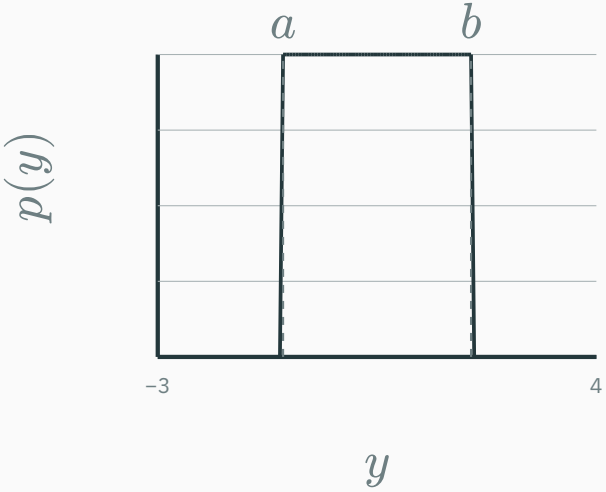
the shaded area is $P(a \leq Y \leq b)$

Uniform distribution: support matters

V VISUAL

$$Y \sim \mathcal{U}(a, b) \quad p(y) = \begin{cases} 1/(b-a) & a \leq y \leq b \\ 0 & \text{otherwise} \end{cases}$$

A likelihood can impose a **hard constraint**.



Checkpoint: uniform support

Q QUESTION

For a Uniform(a, b) observation model, what is the NLL if the target lies outside $[a, b]$?

A. 0

B. A fixed finite penalty

C. $-\log 0 = +\infty$

D. It depends only on σ

Answer: uniform support

A ANSWER

Correct: C — $-\log 0 = +\infty$

Why: The model assigns zero density outside its support, so such a target is impossible under the assumption.

Gaussian distribution

V VISUAL

$$Y \sim \mathcal{N}(\mu, \sigma^2) \qquad p(y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right)$$

Gaussian distribution

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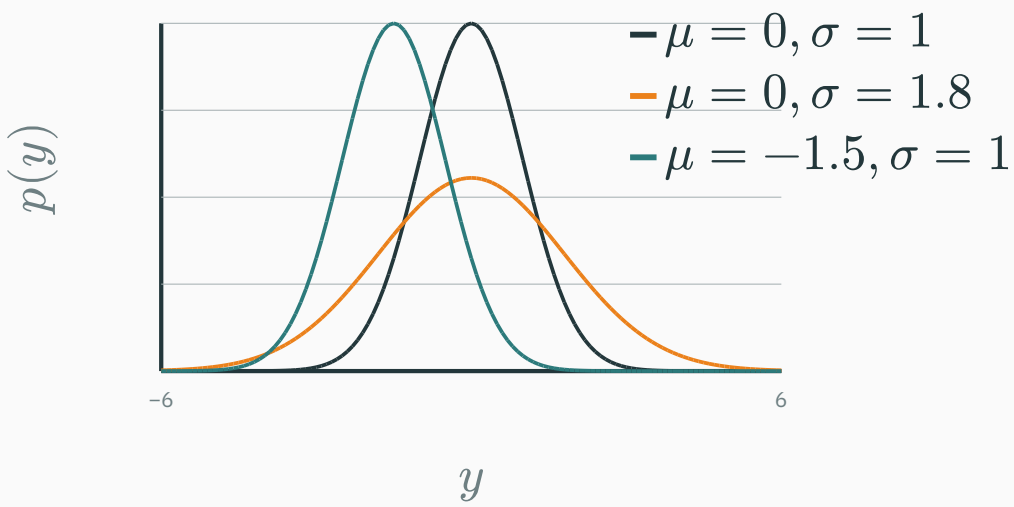

Only two knobs: **location** μ and **scale** σ.

Multivariate Gaussian geometry

V VISUAL

$$\theta \sim \mathcal{N}(\mu, \Sigma) \quad -\log p(\theta) = \frac{1}{2}(\theta - \mu)^\top \Sigma^{-1}(\theta - \mu) + C$$

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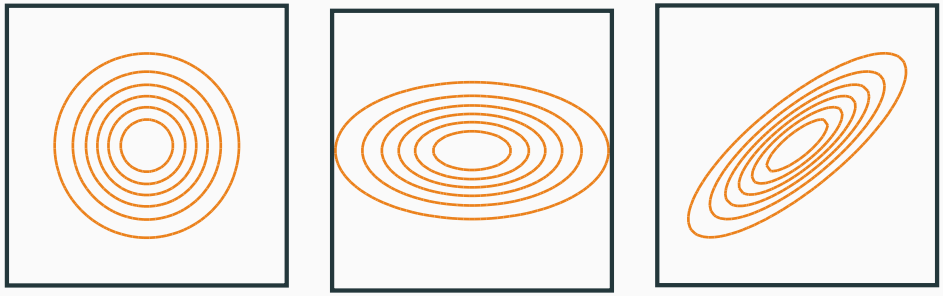
$\Sigma = \tau^2 I$ **diagonal** Σ **full** Σ



Covariance says **which directions in parameter space are plausible.**

Independence turns products into sums

Conditionally independent observations:

$$p(\mathcal{D} \mid \theta) = \prod_{i=1}^{n} p_\theta(y_i \mid x_i)$$

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Why a loss is an **average over examples**, and why minibatches work: $\frac{1}{|B|} \sum_{i \in B} \ell_i$.

Bayes’ rule → likelihood → loss

Bayes' rule and its four pieces

D DERIVATION

Turn the product rule $p(\mathcal{D}, \theta) = p(\mathcal{D} \mid \theta)\, p(\theta) = p(\theta \mid \mathcal{D})\, p(\mathcal{D})$ around:

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$p(\theta \mid \mathcal{D})$	**posterior** — belief about θ *after* seeing data
$p(\mathcal{D} \mid \theta)$	**likelihood** — how well θ explains the data
$p(\theta)$	**prior** — belief about θ *before* data
$p(\mathcal{D})$	**evidence** — the normaliser $\int p(\mathcal{D} \mid \theta) p(\theta)\, \mathrm{d}\theta$

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MLE (now): keep only the likelihood. MAP (later): likelihood $\times$ prior → regularisation.

Maximum likelihood → training losses

Likelihood: one expression, two viewpoints

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The one model $p_\theta(y \mid x)$, read two ways:

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Fix θ, vary y
→ a **distribution** over outcomes.

Fix observed y, vary θ
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Fix θ, vary y
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The likelihood is *not* "the probability that θ is correct."

Coin flips: a likelihood you can compute

D DERIVATION

Ten flips come up $H, H, T, T, T, H, H, T, T, T$ — so $n_H = 4$, $n_T = 6$.

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$$L(\theta) = \prod_i p(x_i \mid \theta) = \theta^{n_H}(1-\theta)^{n_T} = \theta^4(1-\theta)^6$$

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Different θ make the **observed** sequence more or less probable. Which θ makes it most probable?

Coin flips: maximise the (log-)likelihood

D DERIVATION

Logs turn the product into a sum:

$$\mathcal{L}(\theta) = \log L(\theta) = n_H \log \theta + n_T \log(1 - \theta)$$

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Set the derivative to zero:

$$\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}\theta} = \frac{n_H}{\theta} - \frac{n_T}{1-\theta} = 0$$

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The intuitive “fraction of heads” is the maximum-likelihood estimate.

Maximum likelihood estimation

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Interpretation. Choose the parameters under which the observed dataset would be **least surprising**.

From MLE to negative log-likelihood

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$$\arg\max_{\theta} \prod_{i} p_\theta(y_i \mid x_i) = \arg\max_{\theta} \sum_{i} \log p_\theta(y_i \mid x_i)$$

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$$\ell_i(\theta) = -\log p_\theta(y_i \mid x_i)$$

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ML notation — empirical risk:

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Choosing a loss = choosing a probabilistic observation model.

Regression — where does MSE come from?

Regression as signal plus noise

$$y_i = f_\theta(x_i) + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

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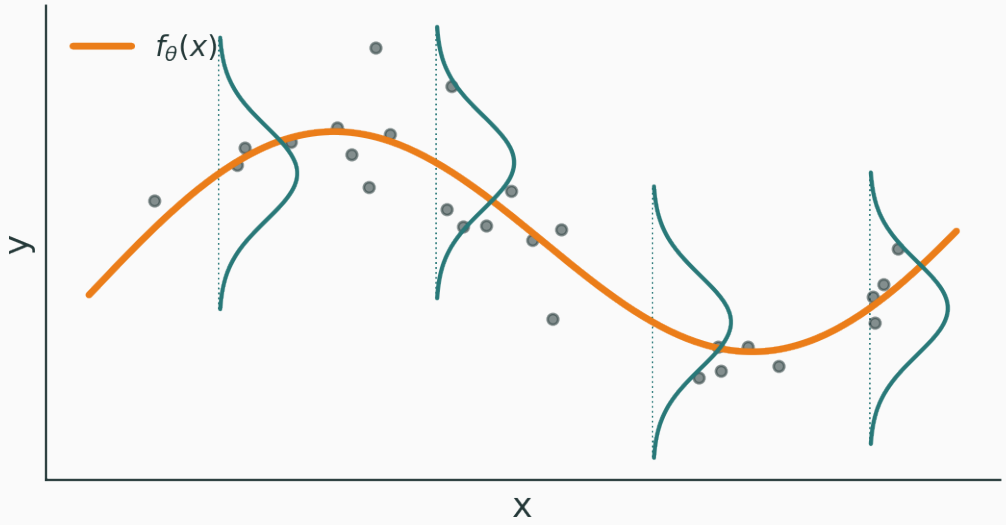
Therefore the conditional output distribution is

$$Y_i \mid x_i, \theta \sim \mathcal{N}\big(f_\theta(x_i), \sigma^2\big)$$

What the assumption means visually

V VISUAL



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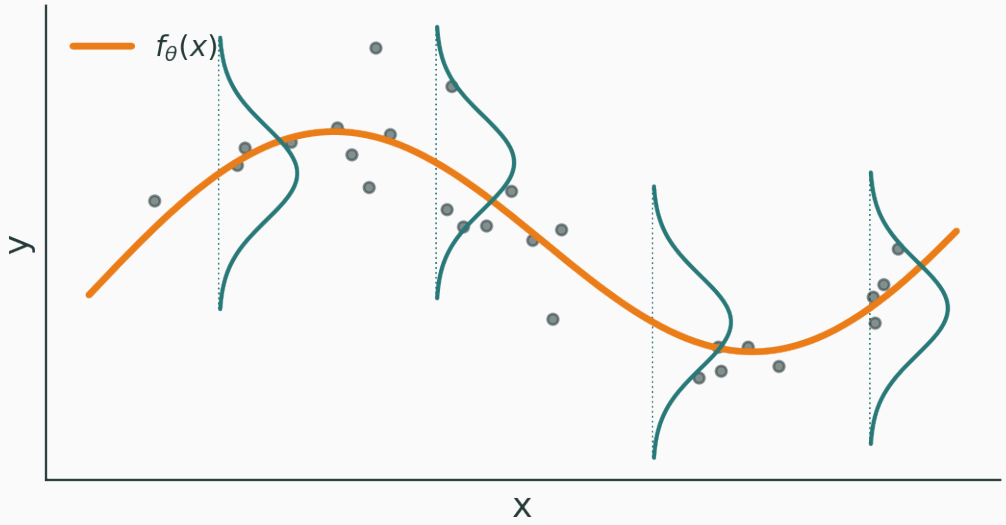

Gaussian noise lives in the **output direction**, conditional on x — a vertical bell at every input.

Gaussian likelihood

D DERIVATION

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Negative log of a single example:

$$-\log p_\theta(y_i \mid x_i) = \frac{(y_i - f_\theta(x_i))^2}{2\sigma^2} + C$$

MSE is Gaussian MLE

Sum over the dataset:

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MSE = Gaussian NLL, up to constants.

What role does σ^2 play?

★ OPTIONAL

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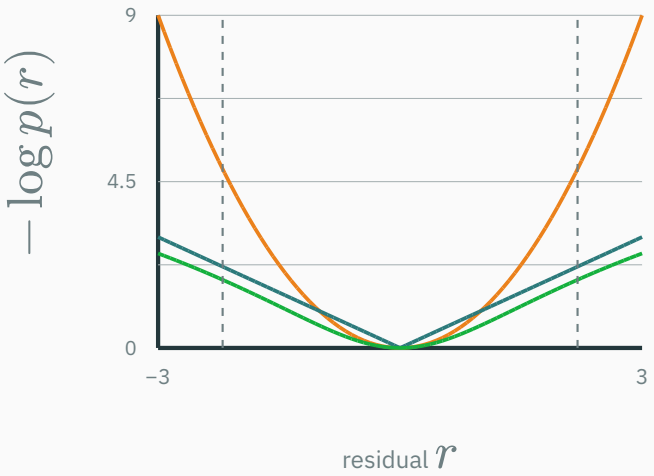
Heteroscedastic regression: let the network output both $(\mu_i, \log\sigma_i^2) = f_\theta(x_i)$.

Different noise, different loss

V VISUAL

Noise model	NLL
Gaussian	r^2
Laplace	$\lvert r \rvert$
Uniform	hard interval
Student-t	robust

$r = y - \hat{y}$ · colours match the curves



Interactive: likelihood → loss

I INTERACTIVE

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/likelihood-to-loss

likelihood-to-loss.html — three synchronized panels: the **noise density** $p(r)$, the **loss** $-\log p(r)$, and **data with a fitted line**.
Controls: Gaussian / Laplace / Uniform / Student-t · noise scale · slope & intercept · **add an outlier** · per-point contributions.

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Ask: which model reacts most to an outlier? why does Uniform give an infinite barrier? which loss is robust?

Classification and cross-entropy

Classification should output probabilities

For K classes:

$$p_\theta(y = k \mid x) \geq 0, \qquad \sum_{k=1}^{K} p_\theta(y = k \mid x) = 1$$

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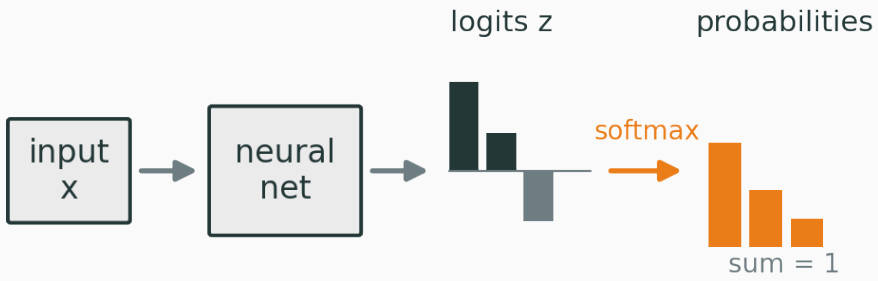

logits $z_k \in \mathbb{R} \rightarrow$ **probabilities** $p_k \in [0, 1]$.

Binary classification: Bernoulli model

$$Y \mid x \sim \text{Bernoulli}(p_\theta(x))$$

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Compact one-line form:

$$p(y \mid x, \theta) = p^y (1-p)^{1-y}$$

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$y = 1 \Rightarrow p(y) = p$

$y = 0 \Rightarrow p(y) = 1 - p$

Bernoulli NLL gives BCE

D DERIVATION

$$-\log p(y \mid x, \theta) = -\log(p^y(1-p)^{1-y})$$

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Binary cross-entropy = Bernoulli NLL.

Bernoulli NLL gives BCE

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$$= -y \log p - (1-y)\log(1-p)$$

Binary cross-entropy = Bernoulli NLL.

Check. True label $y = 1$: if $p = 0.8$ the loss is $-\log 0.8 \approx 0.22$; if $p = 0.2$ it is $-\log 0.2 \approx 1.61$.

Why use logits?

V VISUAL

The network emits an unbounded score $z = f_\theta(x) \in \mathbb{R}$; the **sigmoid** maps it to a probability:

$$p = \sigma(z) = \frac{1}{1 + e^{-z}}$$

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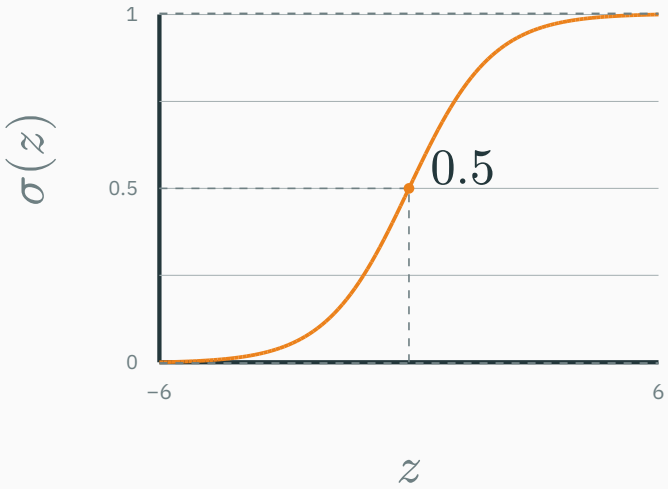
$z \to -\infty : p \to 0$
$z = 0 : p = 0.5$
$z \to +\infty : p \to 1$

In practice sigmoid + BCE are fused for numerical stability.



Logistic regression = Bernoulli MLE

D DERIVATION

Predict each label with the sigmoid of a linear score: $p_i = \sigma(w^T x_i)$, and $Y_i \mid x_i \sim \text{Bernoulli}(p_i)$.

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Exactly the per-example **BCE from two slides ago, summed over the data**. Now differentiate it in w.

The logistic-regression gradient

D DERIVATION

One lemma — the sigmoid's own derivative:

$$\sigma'(z) = \sigma(z)(1 - \sigma(z))$$

The logistic-regression gradient

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$$\frac{\partial J}{\partial w_j} = \sum_{i=1}^{n} (\sigma(w^T x_i) - y_i)\, x_i^j$$

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The gradient is (prediction $-$ target) $\times$ input — the same $p - y$ signal we will meet again for softmax.

Multi-class: categorical model

$$Y \mid x \sim \text{Categorical}(p_1, ..., p_K)$$

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For a one-hot target y:

$$p(y \mid x, \theta) = \prod_{k=1}^{K} p_k^{y_k}$$

From logits to softmax

$$p_k = \frac{e^{z_k}}{\sum_j e^{z_j}}$$

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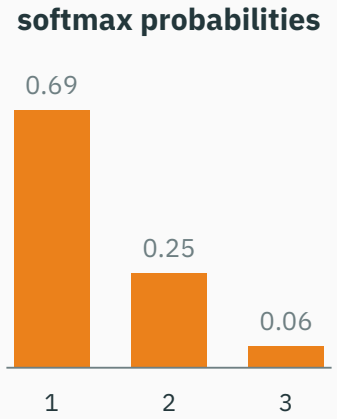

Shift invariance: $\mathrm{softmax}(z) = \mathrm{softmax}(z + c\mathbf{1})$.

Categorical NLL gives cross-entropy

D DERIVATION

$$-\log p(y \mid x, \theta) = -\log \prod_k p_k^{y_k} = -\sum_k y_k \log p_k$$

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Because y is one-hot, only the true class survives:

$$\mathcal{L} = -\log p_{\text{true class}}$$

Confident mistakes are expensive

V VISUAL

$$\ell(p_y) = -\log p_y$$

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Read off the loss at three confidences for the **true** class:

Confident mistakes are expensive

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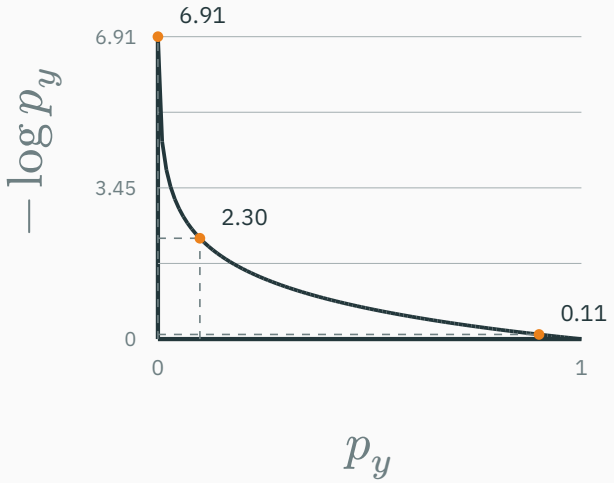
Read off the loss at three confidences for the **true** class:

$p_y = 0.9 \Rightarrow \ell \approx 0.105$
$p_y = 0.1 \Rightarrow \ell \approx 2.303$
$p_y = 0.001 \Rightarrow \ell \approx 6.908$



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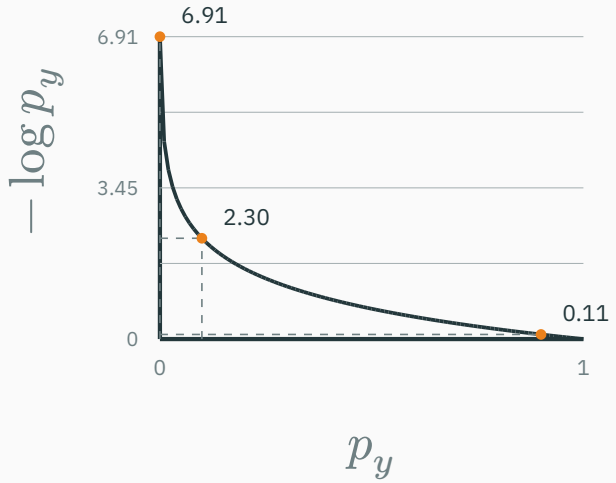

The negative log makes a confident wrong prediction a large loss.

Checkpoint: confident mistakes

Q QUESTION

Why does cross-entropy penalize a confidently wrong prediction strongly?

A. It makes all class probabilities equal

B. The true-class probability is near zero, so $-\log p_y$ is large

C. It adds an L_2 penalty to the logits

D. The gradient is always zero

Answer: confident mistakes

A ANSWER

Correct: B — The true-class probability is near zero

Why: The negative log grows without bound as p_y approaches zero, matching the fact that the model declared the observed class nearly impossible.

A remarkably simple gradient

D DERIVATION

For softmax followed by cross-entropy:

$$\frac{\partial \mathcal{L}}{\partial z_k} = p_k - y_k, \qquad \nabla_z \mathcal{L} = p - y$$

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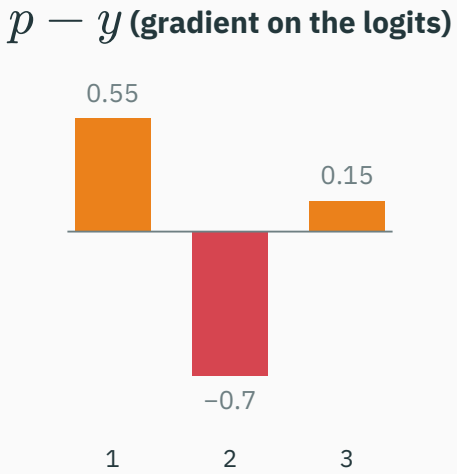

For $p = (0.55, 0.30, 0.15)$, one-hot $y = (0, 1, 0)$: the signal $p - y$ is negative only on the true class — a preview of backprop.

Interactive: logits → softmax → cross-entropy

I INTERACTIVE

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/softmax-cross-entropy

softmax-cross-entropy.html — controls z_1, z_2, z_3, the true class, and temperature T; presets for **correct & confident / correct & unsure / wrong & unsure / wrong & confident**. Shows $z \to \text{softmax}(z/T) \to -\log p_y \to p - y$.

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Ask students to predict the loss before it is revealed.

Why not MSE for classification?

★ OPTIONAL

MSE is not *impossible* — it corresponds to the **wrong observation model** for class indicators.

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MSE assumes $Y_k = p_k + \varepsilon_k, \varepsilon_k \sim \mathcal{N}(0, \sigma^2)$

Reality is $Y \sim \text{Categorical}(p)$

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Cross-entropy also gives **stronger gradients** on confident errors (where MSE's gradient vanishes).

Bayes' rule, directly in classification

Bayes' rule as a classifier

$$p(y = k \mid x) = \frac{p(x \mid y = k)\, p(y = k)}{p(x)}$$

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- $p(x \mid y = k)$ — class-conditional **likelihood**
- $p(y = k \mid x)$ — **posterior** class probability

A 1-D generative classifier

V VISUAL

$$X \mid Y = 0 \sim \mathcal{N}(\mu_0, \sigma_0^2), \quad X \mid Y = 1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$$

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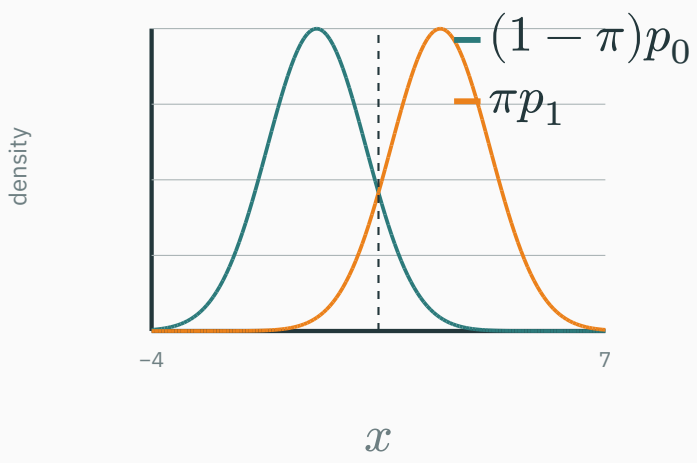
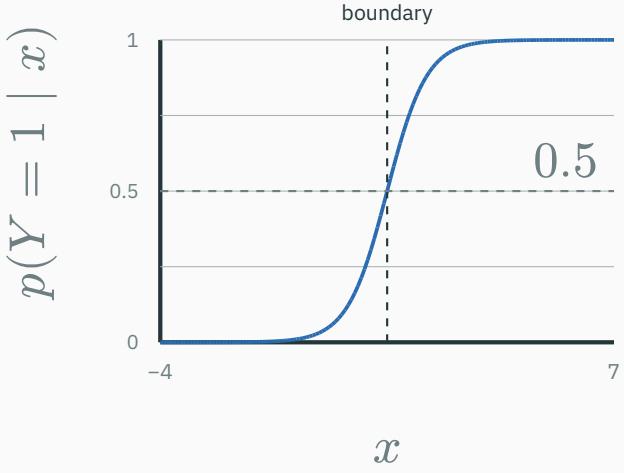


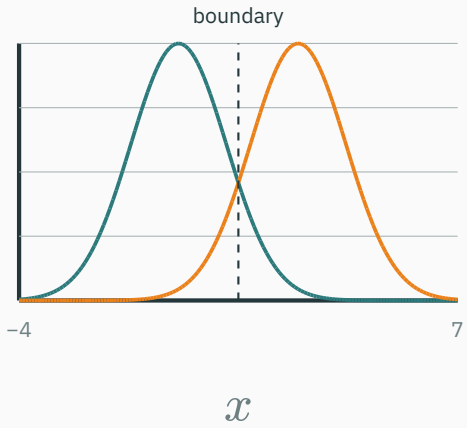


Prior-weighted class densities (left) and the **exact** posterior $p(Y = 1|x)$ (right); the dashed line is the decision boundary, where the posterior crosses 0.5.

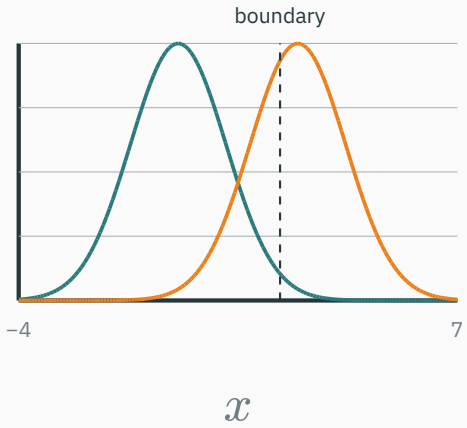
Priors move the decision boundary

V VISUAL

Balanced $p(Y = 1) = 0.5$



Rare $p(Y = 1) = 0.1$



posterior $\propto$ likelihood $\times$ prior

Interactive: Bayes classifier

I INTERACTIVE

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/bayes-classifier

bayes-classifier.html — controls the means μ_0, μ_1, the variances, the **prior** $p(Y = 1)$, and a test point $x^\star$. Shows $p(x \mid Y = 0)$, $p(x \mid Y = 1)$, the posterior $p(Y = 1 \mid x)$, and the decision boundary together.

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Try: balanced → drop the positive prior to 0.1 → why does the boundary move? then widen one variance.

Priors, MAP and regularization

The limitation of MLE

MLE asks only: *how well does θ explain the data?*

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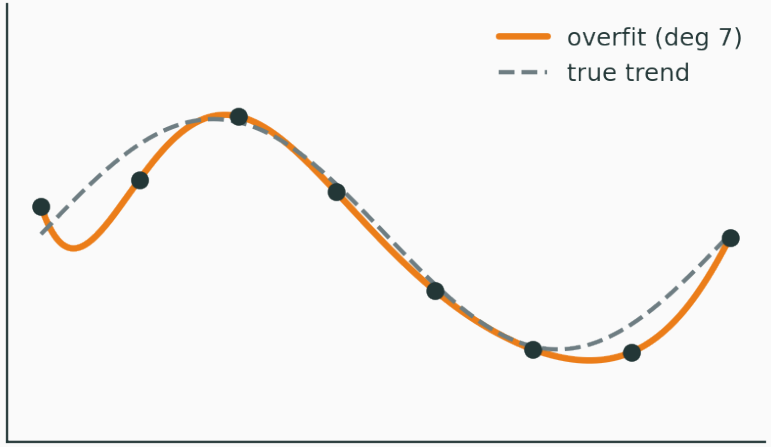

Many parameters fit a small dataset. **Which should we prefer?**

Put a prior over parameters

$\theta \sim p(\theta)$

Before seeing this dataset, which parameter values are plausible?

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- small weights · sparse weights · smooth functions · correlated parameters

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A prior is an **inductive bias**.

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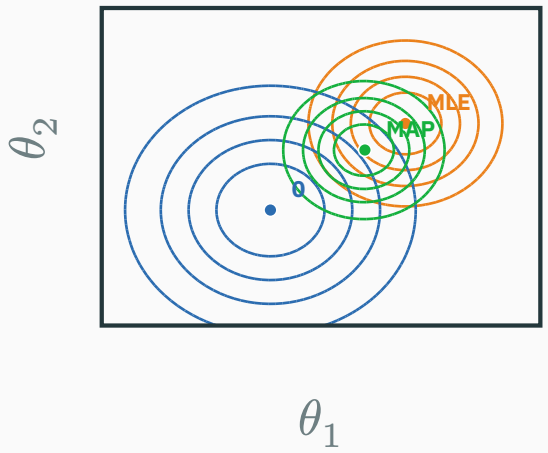

likelihood $\times$ **prior** $\rightarrow$ **posterior** — the MAP is pulled from the MLE toward the prior mean 0.

$p(\mathcal{D}) = \int p(\mathcal{D} \mid \theta) p(\theta)\, \mathrm{d}\theta$ — the evidence (we stop here).

MAP estimation

D DERIVATION

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data loss + regularization

Gaussian prior produces L_2

D DERIVATION

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narrow prior $\Rightarrow$ **stronger** regularization; **broad** prior $\Rightarrow$ **weaker**.

A general Gaussian prior → structured penalty

★ OPTIONAL

For $\theta \sim \mathcal{N}(0, \Sigma)$ the penalty is $\frac{1}{2}\theta^\top \Sigma^{-1}\theta$.

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- unequal variances → penalize parameters **differently**
- correlations → penalize particular **directions**

Laplace prior produces L_1

★ OPTIONAL

$$p(\theta_j) = \frac{1}{2b}\exp\left(-|\theta_j\frac{|}{b}\right) \Rightarrow -\log p(\theta) = \frac{1}{b}\sum_j |\theta_j| + C$$

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Laplace prior $\Rightarrow$ L_1

Heavier tails at zero → **sparsity** (the lasso geometry, if time permits).

Interactive: likelihood, prior, MAP

I INTERACTIVE

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/map-regularization

map-regularization.html — a two-parameter linear model so contours are visible. Shows **likelihood**, **prior**, and **posterior** contours with the **MLE**, **MAP**, and zero points.
Controls: amount of data · noise · prior variance · Gaussian vs Laplace · prior correlation.

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Controls: amount of data · noise · prior variance · Gaussian vs Laplace · prior correlation.

Ask: more data? narrower prior? why does MAP → MLE as data grows? a correlated prior?

Full Bayes and the deep-learning objective

MAP is still one parameter vector

MLE $\hat{\theta}_{\mathrm{MLE}}$ — one point

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Full Bayes keeps the whole posterior $p(\theta \mid \mathcal{D})$ — a **distribution**, not a point.

Posterior predictive distribution

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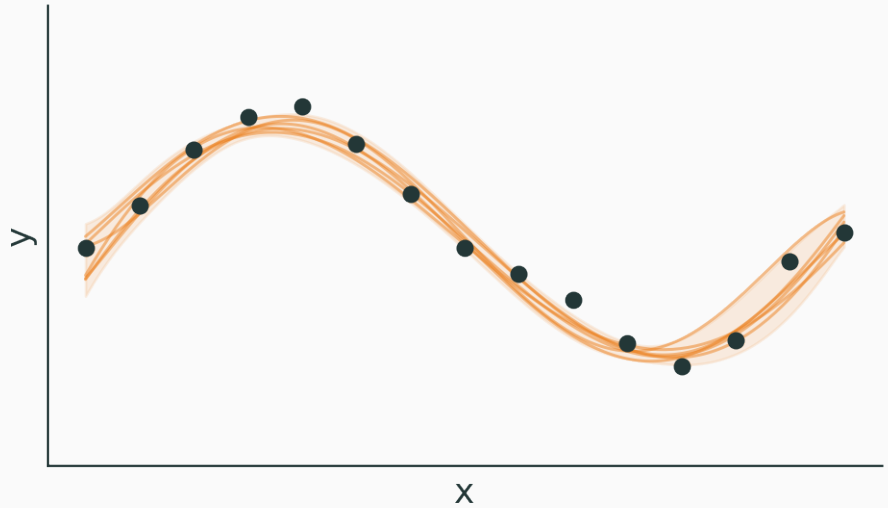

Bayesian prediction **averages over plausible models** — and reports uncertainty.

Why DL usually uses point estimates

★ OPTIONAL

A modern network has millions–billions of parameters, so integrating $p(\theta \mid \mathcal{D})$ is hard.

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- **dropout**-based approximations
- **variational** inference

The standard supervised DL objective

$$\min_{\theta} \underbrace{\frac{1}{|B|} \sum_{i \in B} -\log p_\theta(y_i \mid x_i)}_{\text{likelihood / data fit}} + \underbrace{\lambda \|\theta\|_2^2}_{\text{prior / regularization}}$$

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probabilistic model + neural network + gradient optimization

DL doesn't discard probabilistic ML — it **scales** it with expressive functions and gradient descent.

Retrieval summary

V VISUAL

Gaussian likelihood	$\Rightarrow$ **MSE**
Categorical likelihood	$\Rightarrow$ **cross-entropy**
Gaussian prior	$\Rightarrow L_2$
Laplace prior	$\Rightarrow L_1$
MLE + prior	$\Rightarrow$ **MAP**

Final mental model

See a **loss** → ask *“what likelihood produced it?”*
See a **regularizer** → ask *“what prior produced it?”*
See a **prediction** → ask *“point estimate or posterior?”*

Next: computation graphs, gradients, and backpropagation.